

Operational and Sustainable Development Guidelines for the Broadband Network of the Tusheti Region

1. Strategic Purpose and Operational Context

This regulation has been developed to ensure a 12-month cycle of uninterrupted communications in the Tusheti region. Under conditions of complete winter isolation, when the Abano Pass is closed, the network becomes the only channel for emergency medicine, distance education, and the functioning of government services (Border Police, Protected Areas Agency).

The sustainability philosophy of the Tusheti Development Fund (TDF) is based on the transition from reactive repairs to preventive maintenance. Considering the extreme conditions (altitude above 3000 m, severe frosts, avalanches), operation is built on the principle of “zero failure” during the winter period. Any component that has not passed scheduled inspection in October must be mandatorily replaced before the roads are closed.

2. Technical Architecture and Topology in “Twisted” Terrain Conditions

The network architecture was designed contrary to standard GNCC recommendations. The initial plan to place towers at low elevations near villages failed due to the “twisted” terrain, which blocks line of sight (LoS) and obstructs the Fresnel Zone with evergreen vegetation and snow accumulation up to 1.5 m.

Engineering solution: relocation of node points to dominant elevations (ridge lines) and the use of towers no lower than 4 meters. This ensures the clearance of the first Fresnel Zone above snow cover and tree canopies.

Key Nodes and Backbone Structure

Node	Type	Equipment	Specifics / Risks
Ruispiri	Output Point	AF-5x, RD-34db, RAD-3RD, RB2011	Main uplink to Freenet backbone
Abano	Main Bridge	AF-5x, RD-34db, RAD-3RD, AF-5G-OMT-S45	High lightning risk zone (ferrites in rock formations)
Diklo	Basic Point	AF-5x, PBE-5AC-620, LBE-5AC-16-120	Central distribution hub
Chiglaurta	Basic Point	PBE-5AC-620, RB750P-PBr2	Main relay point

Makratela	Repeater	PBE-5AC-620, LBE-5AC-16-120	Coverage for Indurta and Parsma
Dochu	Node Point	PBE-5AC-620, Sector Am-5G19-120	Gometari node
Koklata	Basic Point	PBE-5AC-620, LBE-5AC-16-120	Extreme winds and icing
Dano	Node Point	PBE-5AC-620, Sector Antennas	Coverage for Chesho and Kvavlo

Note: To protect links from wind and snow loads on backbone sections (Abano–Diklo), the installation of RocketDish Radome (RAD-3RD) radio-transparent covers and Antenna Conversion Kit (AF-5G-OMT-S45) sets is mandatory for polarization optimization.

3. Maintenance Regulations for Solar Power Generation and Energy Storage Systems

Winter operation in Tusheti is limited by a fixed insolation angle (53°). During this period, panel efficiency drops to a critical level at which generation cannot simultaneously power active equipment and charge the batteries.

Technical Power Configuration

- **Power upgrade:** Transition from 120W systems to 200W/12V per tower (installation of four 100W panels connected in series-parallel configuration to achieve 24V).
- **Injectors:** Use of POE-BOX8-G for stable power supply to the entire tower infrastructure.
- **Batteries:** Capacity no less than 100–105 Ah (6V/12V cells).
- **Controllers:** Nippotec or Andeli systems.

Battery Maintenance Protocol

- **October inspection:** Measurement of residual capacity at all nodes.
- **Rotation algorithm:** If battery capacity at the Kehi or Koklata nodes falls below 85%, immediate rotation is required. Worn batteries are moved to less critical accessible nodes, while fresh reserve units from Alvani are installed at high-altitude points.
- **Contacts:** Before winter, mandatory tightening of terminals and treatment with anti-corrosion compounds.

4. Minimization of Geophysical and Environmental Risks

Lightning Protection (Abano Case)

The experience of the operator Mobitel demonstrated that standard grounding is ineffective on peaks with high sulfide and ferrite content.

- **Protocol:** Use of specialized Andeli lightning protection systems combined with grounding compliant with ISO 9001:2008 standards. This is critically important for the Abano node.

Snow and Wind Load

- **Mechanical requirements:** Towers and shelters must withstand snow pressure up to 1.5 m thick. Inspection of anchor tension and guy wires after the thawing period (May) is mandatory.
- **Monitoring:** Installation of IP cameras with night vision at the Kehi and Omalo nodes. Cameras must focus on solar panels and cover the visibility sectors of Kesselo / Gometari / Bochora to assess snow accumulation.

5. Operational Procedures and Work Schedule (SOP)

7-Day Field Deployment / Maintenance Plan

1. **Day 1:** Preparation in Ruispiri. Equipment loading. Delivery to Diklo (truck).
2. **Day 2:** Ascent to Abano Pass (SUV Delica). Installation/alignment of the link to Ruispiri (Alazani Valley). Descent to Diklo.
3. **Day 3:** Ascent to Diklo Mountain (on horseback). Installation/alignment of the bridge with Abano.
4. **Day 4:** Transportation to Dochu / Koklata. Delivery of equipment to Dochu elevation. Traffic testing.
5. **Day 5:** Ascent to Koklata Mountain. Alignment of the Dochu–Koklata link. Group A — installation of client routers; Group B — preparation in Diklo.
6. **Day 6:** Ascent to Dano. Integration with Diklo. Group B — installation in Dano and Diklo. Group A — final retest of the Ruispiri–Diklo–Koklata route.
7. **Day 7:** Operations in the Dartlo and Omalo gorges. Mass connection of guesthouses.

Logistics

- **Heavy equipment (batteries, panels):** 4x4 cargo vehicles.
- **Engineering teams:** Mitsubishi Delica 4x4.
- **High-altitude points (Kehi, Diklo, Makratela):** Use of horses — the only delivery method under conditions of blocked mountain trails.

Personnel reserve: Training of at least two local specialists (LAN cabling, crimping, IP configuration, basic troubleshooting in power cabinets).

6. Sustainable Development and Management Model (TDF)

The network operates as a Community-led ISP under the auspices of the Tusheti Development Fund.

Financial and Legal Parameters

- **Profit limitation:** No more than 10% above operational (OPEX) and capital (CAPEX) expenditures.

- **Reinvestment:** All income is directed toward modernization of nodes and expansion of coverage to hard-to-reach villages (Kumelaurta, Indurta).
- **External contribution:** Partnership with Freenet (Ruispiri node) provides a 100 Mbps channel (estimated contribution — 3200 GEL/month).
- **Obligations:** Non-discriminatory access at all points of Tusheti where LoS is available.

Priority Subscriber Groups

1. **Guesthouses (SME):** The main source of income for the self-sustainability of the fund (more than 40 facilities in Omalo, Dartlo, Shenako).
2. **Government sector:** Border Police and Protected Areas Agency. Their transition to the TDF network ensures a guaranteed funding stream for 12-month operational support.
3. **Local population:** Social tariff plans to ensure communications during the winter period.

Compliance with this regulation is the only guarantee of infrastructure functionality under the extreme high-mountain conditions of Tusheti.